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Inventor(s) Fredricks; Eric R. et al.

SYSTEM AND METHOD FOR CONTROLLING THE OPERATION OF AN AGRICULTURAL HARVESTER

Abstract

A harvesting implement of an agricultural harvester includes a frame configured to be coupled to a harvester base vehicle. Additionally, the implement includes a cutter bar configured to sever crop material from the field and a reel rotatably coupled to the frame and configured to direct the crop material toward the cutter bar. Furthermore, the implement includes a sensor assembly including an arm coupled to the frame. The arm is configured to rotate relative to the frame between a stored position and an open position in which the arm is configured to contact the surface of the field. Additionally, the implement includes a biasing element for biasing the arm toward the field. Moreover, the implement includes an electrically activatable actuator coupled to the arm and configured to rotate the arm between the stored and open positions and apply a load to prevent the arm from rotating.

Inventors: **Fredricks; Eric R. (Lancaster, PA), Gessel; James M. (Geneseo, IL)**

Applicant: **CNH Industrial America LLC** (New Holland, PA)

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Background/Summary

FIELD OF THE INVENTION

[0001] The present disclosure generally relates to agricultural harvesters and, more particularly, to systems and methods for controlling the operation of an agricultural harvester.

BACKGROUND OF THE INVENTION

[0002] A harvester is an agricultural machine used to harvest and process crops. For instance, a forage harvester may be used to cut and comminute silage crops, such as grass and corn. Similarly, a combine harvester may be used to harvest grain crops, such as wheat, oats, rye, barely, corn, soybeans, and flax or linseed. In general, the objective is to complete several processes, which traditionally were distinct, in one pass of the machine over a particular part of the field. In this regard, most harvesters are equipped with a detachable harvesting implement, such as a header, which cuts and collects the crop from the field and feeds it to the harvester base vehicle for further processing.

[0003] The operator of the harvester typically adjusts the height of the header during harvesting operations. As such, the header includes one or more height sensor assemblies often in the form of contact sensors with rotatable sensor arms configured to engage the surface of the field and, thus, indicate the height of the header to the operator. However, the sensor arms often require manual adjustments by the operator, which may interrupt harvesting operations.

[0004] Accordingly, an improved system and method for controlling the operation of an agricultural harvester would be welcomed in the technology.

SUMMARY OF THE INVENTION

[0005] Aspects and advantages of the technology will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the technology.

[0006] In one aspect, the present subject matter is directed to a harvesting implement. The harvesting implement includes a frame configured to be coupled to a harvester base vehicle. Additionally, the harvesting implement includes a cutter bar configured to sever crop material from a field. Furthermore, the harvesting implement includes a reel rotatable coupled to the frame and configured to direct crop material toward the cutter bar. Moreover, the harvesting implement includes an implement height sensor assembly including an arm coupled to the frame. The arm is configured to rotate relative to the frame between a stored position and an open position in which the arm is configured to contact a surface of the field as the harvesting implement is moved through the field. Furthermore, the harvesting implement includes a biasing element configured to bias the arm of the implement height sensor toward the surface of the field when the arm is in the open position. Additionally, the harvesting implement includes an electrically activatable actuator coupled to the arm of the implement height sensor assembly and configured to rotate the arm between the stored position and the open position and apply a load to prevent the arm from rotating from the stored position or the open position.

[0007] In another aspect, the present subject matter is directed to a system for controlling the operation of an agricultural harvester. The system includes a harvesting implement pivotably coupled to a harvester base vehicle. The harvesting implement is configured to be pivoted such that a height of the harvesting implement above a surface of a field is adjusted. Additionally, the harvesting implement includes an implement height sensor assembly including an arm coupled to the harvesting implement. The arm is configured to rotate relative to the harvesting implement between a stored position and an open position in which the arm is configured to contact a surface of the field as the harvesting implement is moved through the field. Furthermore, the system includes an electrically activatable actuator coupled to the arm of the implement height sensor assembly and configured to rotate the arm between the stored position and the open position and apply a load to prevent the arm from rotating from the stored position or the open position. Moreover, the harvesting implement includes a computing system. The computing system is configured to receive an input to rotate the arm between the stored position and the open position. Additionally, the computing system is configured to control the operation of the electrically activatable actuator to rotate the arm between the stored position and the open position based on the received input and, once the arm is in position, prevent the arm from rotating.

[0008] In another aspect, the present subject matter is directed to a method for controlling the operation of an agricultural harvester. The agricultural harvester includes a harvesting implement coupled to a harvester base vehicle. The method includes receiving, with a computing system, an input to rotate an arm of an implement height sensor assembly relative to the harvesting implement between a stored position and an open position in which the arm is configured to contact a surface of the field as the harvesting implement is moved through the field. Additionally, the method includes controlling, with the computing system, an operation of an electrically activatable actuator coupled to the arm of the implement height sensor assembly to rotate the arm between the stored position and the open position based on the received input and, once the arm is in position, prevent the arm from rotating.

[0009] These and other features, aspects and advantages of the present technology will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the technology and, together with the description, serve to explain the principles of the technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A full and enabling disclosure of the present technology, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0011] FIG. **1** illustrates a partial sectional side view of one embodiment of an agricultural harvester in accordance with aspects of the present subject matter;

[0012] FIG. **2** illustrates a perspective view of one embodiment of a harvesting implement of an agricultural harvester in accordance with aspects of the present subject matter;

[0013] FIG. **3** illustrates a schematic view of one embodiment of a system for controlling the operation of an agricultural harvester in accordance with aspects of the present subject matter;

[0014] FIG. **4** illustrates a flow diagram of one embodiment of control logic for controlling the operation of an agricultural harvester in accordance with aspects of the present subject matter; and

[0015] FIG. **5** illustrates a flow diagram of one embodiment of a method for controlling the operation of an agricultural harvester in accordance with aspects of the present subject matter.

[0016] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present technology.

DETAILED DESCRIPTION OF THE DRAWINGS

[0017] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0018] In general, the present subject matter is directed to systems and methods for controlling the operation of an agricultural harvester. Specifically, in several embodiments, the agricultural harvester includes a harvesting implement that includes a frame configured to be coupled to the harvester base vehicle of the agricultural harvester. The harvesting implement also includes a cutter bar configured to sever the crop material from the field, and a reel rotatably coupled to the frame of the harvesting implement configured to direct the crop material toward the cutter bar to be severed. Additionally, the harvesting implement includes one or more implement height sensor assemblies, each of which includes a sensor arm. The sensor arm(s) is coupled to the frame of the harvesting implement and configured to rotate relative to the frame between a stored position and an open position. When the sensor arm(s) is in the open position, the sensor arm(s) is configured to contact the surface of the field as the harvesting implement is moved through the field.

[0019] Additionally, an electrically activatable actuator, such as an electromagnetic actuator, is coupled to the sensor arm(s) and configured to rotate the sensor arm(s) between the stored position and the open position. Furthermore, once the sensor arm(s) is in position, the electrically activatable actuator is configured to apply a load (e.g., electromagnetic load) to prevent the sensor arm(s) from rotating from the stored position or the open position. Furthermore, in several embodiments, a computing system of the disclosed system is configured to control the operation of the electrically activatable actuator to rotate the arm between the stored position and the open position based on a received input, such as an operator input, to rotate the arm between the stored position and the open position.

[0020] Dynamically adjusting the load applied to the sensor arm(s) as the agricultural harvester traverses the field improves the operation of the agricultural harvester. In particular, the sensor arm(s) is oftentimes easily pivoted or rotated when the sensor arm(s) engages the field surface and, thus, a pressure or load must be applied to the sensor arm(s), which is typically accomplished with a spring that must be manually adjusted and/or using heavy materials for the sensor arm(s). However, these practices of applying loads to the sensor arm(s) does not allow for dynamic adjustment of the amount of the load that is being applied to the sensor arm(s) to ensure that the sensor arm(s) maintains contact with the field surface during harvesting operations. Dynamic adjustment of the amount of load being applied to the sensor arm(s) is useful when conditions of the field vary, such as soil compaction levels and/or the roughness of the field surface, and/or the ground speed of the harvester base vehicle, since such conditions of the field may affect the rotation of the sensor arm(s) when the sensor arm(s) engages the field surface. Such dynamic adjustment prevents the operator from having to manually adjust the pressure or load being applied to the sensor arm(s) as the field conditions change.

[0021] Referring now to the drawings, FIG. **1** illustrates a partial sectional side view of the agricultural harvester **10**. In general, the harvester **10** may be configured to travel across a field in a forward direction of travel (e.g., as indicated by arrow **12**) to harvest a crop **14**. While traversing the field, the harvester **10** may be configured to process and store the harvested crop within a crop tank **16** of the harvester **10**. Furthermore, the harvested crop may be unloaded from the crop tank **16** for receipt by the crop receiving vehicle (not shown) via a crop discharge tube **18** of the harvester **10**.

[0022] As shown, in one embodiment, the harvester **10** may be configured as an axial-flow type combine in which the harvested crop is threshed and separated while being advanced by and along a longitudinally arranged rotor **20**. However, it should be appreciated that, in alternative embodiments, the harvester **10** may have any other suitable harvester configuration, such as a traverse-flow type configuration.

[0023] The harvester **10** may include a harvester base vehicle **52** and a harvesting implement, such as a header **34**, coupled to the harvester base vehicle **52**. The harvester base vehicle **52** may include chassis or main frame **22** configured to support and/or couple to various components of the harvester **10**. For example, in several embodiments, the harvester base vehicle **52** may include a pair of driven, ground-engaging front wheels **24** and a pair of steerable rear wheels **26** that are coupled to the frame **22**. As such, the wheels **24**, **26** may be configured to support the harvester **10** relative to the ground and move the harvester **10** in the forward direction of travel **12**.

[0024] Furthermore, the harvester base vehicle **52** may include an operator's platform **28** having an operator's cab **30**, a crop processing system **32**, the crop tank **16**, and the crop discharge tube **18** that are supported by the frame **22**. As will be described below, the crop processing system **32** may be configured to perform various processing operations on the harvested crop as the crop processing system **32** operates to transfer the harvested crop between the header **34** and the crop tank **16**. Furthermore, the harvester base vehicle **52** may include an engine **36** and a transmission **38** mounted on the frame **22**. The transmission **38** may be operably coupled to the engine **36** and may provide variably adjusted gear ratios for transferring engine power to the wheels **24** via a drive axle assembly (or via axles if multiple drive axles are employed).

[0025] Moreover, as shown in FIG. **1**, the header **34** and an associated feeder **40** of the crop processing system **32** may extend forward of the frame **22** and may be pivotally secured thereto for generally vertical movement. In general, the feeder **40** may be configured to serve as a support structure for the header **34**. As shown in FIG. **1**, the feeder **40** may extend between a front end **42** coupled to the header **34** and a rear end **44** positioned adjacent to a threshing and separating assembly **46** of the crop processing system **32**. As is generally understood, the rear end **44** of the feeder **40** may be pivotally coupled to a portion of the harvester **10** to allow the front end **42** of the feeder **40** and, thus, the header **34** to be moved upward and downward along a vertical direction (e.g., as indicated by arrow **48**) relative to a field surface **19** to set the desired harvesting or cutting height for the header **34**. For example, as shown, in one embodiment, the harvester **10** may include a height actuator **50** configured to adjust the height of the header **34** relative to the ground. As such, the height actuator **50** may correspond to a fluid-driven actuator, such as a hydraulic or pneumatic cylinder, an electric linear actuator, or any other type of suitable actuator.

[0026] In some embodiments, the header **34** includes a reel **92** rotatably coupled to a reel frame **94** which is, in turn, coupled to a frame **82** of the header **34**. The reel **92** is generally configured to contact crop material before a cutter bar **96** of the header **34**. For instance, the reel **92** may include tines and/or the like such that, when crop materials contact the reel **92**, the crop materials may be oriented into a substantially uniform direction and guided toward the cutter bar **96**. The vertical positioning of the reel **92** (e.g., relative to the ground and/or frame **22**) may be adjusted by a reel actuator **41** coupled between the reel frame **94** and the feeder **40**. For instance, the reel actuator **41** may be a cylinder which is extendable and retractable to adjust a vertical position of the reel **92**.

[0027] As the harvester **10** is propelled forwardly over a field with standing crop, the crop material is directed towards the cutter bar **96** by the reel **92**. The crop material is severed from the stubble by the cutter bar **96** at the front of the header **34** and delivered by a header auger **54** to the front end **42** of the feeder **40**, which supplies the cut crop to the threshing and separating assembly **46**. As is generally understood, the threshing and separating assembly **46** may include a cylindrical chamber **56** in which the rotor **20** is rotated to thresh and separate the crop received therein. That is, the crop is rubbed and beaten between the rotor **20** and the inner surfaces of the chamber **56**, whereby the grain, seed, or the like, is loosened and separated from the straw.

[0028] The harvested crop that has been separated by the threshing and separating assembly **46** may fall onto a crop cleaning assembly **58** of the crop processing system **32**. In general, the crop cleaning assembly **58** may include a series of pans **60** and associated sieves **62**, with the separated crop material being spread out via oscillation of the pans **60** and/or sieves **62** and eventually falling through apertures defined in the sieves **62**. Additionally, a cleaning fan **64** may be positioned adjacent to one or more of the sieves **62** to provide an air flow through the sieves **62** that removes chaff and other impurities from the crop material. For instance, the fan **64** may blow the impurities off of the crop material for discharge from the harvester **10** through the outlet of a straw hood **66** positioned at the back end of the harvester **10**. The cleaned crop material passing through the sieves **50** may then fall into a trough of an auger **68**, which may be configured to transfer the crop material to an elevator **70** for delivery to the crop tank **16**.

[0029] The configuration of the harvester **10** described above and shown in FIG. **1** is provided only to place the present subject matter in an exemplary field of use. Thus, the present subject matter may be readily adaptable to any manner of harvester configuration.

[0030] Furthermore, the harvester **10** may include one or more field condition sensors **72** coupled thereto and/or supported thereon. The field condition sensor(s) **72** is configured to generate data indicative of one or more conditions of the field across which the harvester **10** is traversing, such as data indicative of the compaction level of soil of the field and/or data indicative of the roughness of the surface of the field. As will be described below, such data may subsequently be used to control the operation of an electrically activatable actuator **120** to rotate one or more sensor arms **106** (FIG. **2**) of one or more implement height sensor assemblies **102** (FIG. **2**).

[0031] In general, the field condition sensor(s) **72** may correspond to any suitable sensor(s) configured to generate data indicative of the compaction level of the soil of the field and/or data indicative of the roughness of the surface of the field. In several embodiments, the field condition sensor(s) **72** may be configured as an imaging device(s), such as a light detection and ranging (LiDAR), camera(s), and/or the like configured to depict a portion of the field within a field(s) of view **76** of the field condition sensor(s) **72**. Furthermore, in alternative embodiments, the field condition sensor(s) **72** may be configured as any other suitable sensor(s) for generating data indicative of the compaction level of the soil of the field and/or the roughness of the surface of the field.

[0032] Furthermore, the harvester **10** may include any number of field condition sensors **72** provided at any suitable location that allows data indicative of the compaction level of the soil of the field and/or the roughness of the surface of the field to be generated. In this respect, FIG. **1** illustrates an example location for mounting the field condition sensor(s) **72** for generating data indicative of the compaction level of the soil of the field and/or the roughness of the surface of the field. For example, in embodiments in which the field condition sensor(s) **72** is configured as an imaging device(s), the field condition sensor(s) **72** may be mounted on the operator's cab **30** of the harvester **10** such that the field condition sensor(s) **72** has the field(s) of view **76** directed forward of the of the harvester **10**. However, in alternative embodiments, the field condition sensor(s) **72** may be installed at any other suitable location(s) that allows the device(s) to generate data indicative of the compaction level of the soil of the field and/or the roughness of the surface of the field.

[0033] Additionally, the harvester **10** may include one or more vehicle speed sensors **74** coupled thereto and/or supported thereon. The vehicle speed sensor(s) **74** is configured to generate data indicative of the ground speed of the harvester base vehicle **52** as the harvester **10** traverses the field. As will be described below, such data may subsequently be used to control the operation of the electrically activatable actuator **120** to rotate the sensor arm(s) **106** of the implement height sensor assembly(ies) **102**.

[0034] In general, the vehicle speed sensor(s) **74** may correspond to any suitable sensor(s) configured to generate data indicative of the ground speed of the harvester base vehicle **52**. In several embodiments, the vehicle speed sensor(s) **74** may be configured as an accelerometer and/or the like. Furthermore, in alternative embodiments, the vehicle speed sensor(s) **74** may be configured as any other suitable sensor(s) for generating data indicative of the ground speed of the harvester base vehicle **52**.

[0035] Furthermore, the harvester **10** may include any number of vehicle speed sensors **74** provided at any suitable location that allows data indicative of the ground speed of the harvester base vehicle **52** to be generated.

[0036] Referring now to FIG. **2**, a perspective view of one embodiment of the header **34** is illustrated in accordance with aspects of the present subject matter. As shown in FIG. **2**, one or more height sensor assemblies **102** may be provided in operative association with the header **34**. As indicated above, the header **34** may be moved upward and downward along the vertical direction **48** relative to the field surface **19** to set the desired harvesting or cutting height. In several embodiments, the height sensor assembly(ies) **102** may be configured to detect a parameter indicative of the height or distance (e.g., as indicated by arrow **84** in FIG. **2**) between the header **34** (e.g., a bottom **86** of the header **34**) and the field surface **19**.

[0037] In several embodiments, the height sensor assembly(ies) **102** may be configured as a contact-based sensor assembly(ies). For example, the height sensor assembly(ies) **102** may include a rotary sensor **104** (e.g., a rotary potentiometer or a magnetic rotary sensor) coupled to the frame **82** of the header **34** and a feeler or sensor arm **106** having a first end **108** pivotally associated with the rotary sensor **104** and an opposed second end **110** configured to engage the field surface **19**. As such, when the height **84** of the header **34** is adjusted, the sensor arm **106** may pivot or rotate relative to the rotary sensor **104** and the frame **82** of the header **34** between a stored position **88**A, in which the sensor arm **106** is fully retracted, and an open position **88**B, in which the second end **110** of the sensor arm **106** is configured to engage the field surface **19**. Additionally, the height sensor assembly(ies) **102** may include a biasing element **78**, such as a spring, configured to bias the sensor arm **106** toward the field surface **19** when the sensor arm **106** is in the open position **88**B. The rotary sensor **104** may, in turn, detect the pivotal or rotational motion of the sensor arm **106**, with such pivotal movement being indicative of the height **84** of the header **34**. Furthermore, the rotary sensor **104** may be configured to generate data indicative of the position of the sensor arm **106** relative to the frame **82** of the header **34**. As will be described below, such data may be used by one or more computing systems to control the operation of the electrically activatable actuator **120** to rotate the sensor arm(s) **106**.

[0038] For example, in one embodiment, the height sensor assembly(ies) **102** may include a sensor arm shaft **112** that extends between a first end **114** and a second end **116** in a lateral direction (as indicated by arrow **98**) perpendicular to the vertical direction **48**. The sensor arm shaft **112** may be coupled to the frame **82** of the header **34** at its first and second ends **114**, **116** and may be configured to rotate about an axis of rotation **90** defined between the first and second ends **114**, **116** parallel to the lateral direction **98**. The sensor arm(s) **106** of the height sensor assembly(ies) **102** may be fixed to the sensor arm shaft **112**. In this respect, as the sensor arm shaft **112** rotates about the axis of rotation **90**, the sensor arm(s) **106** rotates with the sensor arm shaft **112** between the stored position **88**A and the open position **88**B.

[0039] In general, the sensor arm(s) **106** of the height sensor assembly(ies) **102** is easily pivoted or rotated when the sensor arm(s) **106** engages the field surface **19** and, thus, a pressure or load must be applied to the sensor arm(s) **106**, which is typically accomplished with the biasing element **78**, such as a spring or heavy materials for the sensor arm(s) **106**. However, these practices of applying loads to the sensor arm(s) **106** do not allow for dynamic adjustment of the amount of the load that is being applied to the sensor arm(s) **106** to ensure that the sensor arm(s) **106** maintains contact with the field surface **19**. Dynamic adjustment of the amount of load being applied to the sensor arm(s) **106** is useful when conditions of the field vary, such as soil compaction levels and/or the roughness of the field surface **19**, and/or the ground speed of the harvester base vehicle **52**, since such conditions of the field may affect the rotation of the sensor arm(s) **106** when the sensor arm(s) **106** engages the field surface **19**. For example, increased roughness of the field surface **19**, increased density of soil compaction, and/or increased speed of the harvester base vehicle **52** is directly proportional to increased pivoting of the sensor arm(s) **106** when the same pressure or load is applied to the sensor arm(s) **106**.

[0040] As such, the header **34** may include one or more electrically activatable actuators **120**. The electrically activatable actuator(s) **120** may be coupled to the sensor arm(s) **106** of the height sensor assembly(ies) **102** and configured to pivot or rotate the sensor arm(s) **106** between the stored position **88**A and the open position **88**B. Additionally, the electrically activatable actuator(s) **120** may be configured to apply a pressure or load to the sensor arm(s) **106** to limit or prevent the sensor arm(s) **106** from rotating from the stored position **88**A or the open position **88**B.

[0041] In some embodiments, the electrically activatable actuator(s) **120** may be configured as an electromagnetic actuator(s) **118**, such as an electric motor(s). The electromagnetic actuator(s) **118** are configured to apply an electromagnetic load to rotate the sensor arm(s) **106** between the stored position **88**A and the open position **88**B. For example, as shown in FIG. **2**, the electromagnetic actuator(s) **118** may be coupled to the sensor arm shaft **112** of the header **34** and configured to apply an electromagnetic load such that the sensor arm shaft **112** and, thus, the sensor arm(s) **106**, is rotated about the axis of rotation **90**. Additionally, when the sensor arm(s) **106** is in the open position **88**B, the electromagnetic load applied by the electromagnetic actuator(s) **118** counteracts a pressure or load applied to the sensor arm(s) **106** by the field surface **19** to limit or prevent the sensor arm(s) **106** from rotating. As will be described below, one or more computing systems may adjust a magnitude of the electromagnet load applied by the electromagnetic actuator(s) **118**.

[0042] Referring now to FIG. **3**, a schematic view of one embodiment of a system **200** for controlling the operation of an agricultural harvester is illustrated in accordance with aspects of the present subject matter. In general, the system **200** will be described herein with reference to the agricultural harvester **10** described above with reference to FIGS. **1** and **2**. However, the disclosed system **200** may generally be utilized with agricultural harvester having any other suitable harvester configuration.

[0043] As shown in FIG. **3**, the system **200** generally includes one or more components of the harvester **10**. For example, in the illustrated embodiment, the system **200** includes the field condition sensor(s) **72**, the vehicle speed sensor(s) **74**, the rotary sensor(s) **104**, and the electromagnetic actuator(s) **118** of the header **34**.

[0044] Moreover, the system **200** includes a computing system **210** communicatively coupled to one or more components of the harvester **10** and/or the system **200** to allow the operation of such components to be electronically or automatically controlled by the computing system **210**. For instance, the computing system **210** may be communicatively coupled to the field condition sensor(s) **72** via a communicative link **202**. As such, the computing system **210** may be configured to receive data from the field condition sensor(s) **72**, such as data indicative of the compaction level of the soil of the field and/or data indicative of the roughness of the surface of the field. Furthermore, the computing system **210** may be communicatively coupled to the vehicle speed sensor(s) **74** via the communicative link **202**. In this respect, the computing system **210** may be configured to receive data from the vehicle speed sensor(s) **74**, such as data indicative of the ground speed of the harvester base vehicle **52**. Additionally, the computing system **210** may be communicatively coupled to the rotary sensor(s) **104** via the communicative link **202**. In this respect, the computing system **210** may be configured to receive data from the rotary sensor(s) **104**, such as data indicative of the position(s) of the sensor arm(s) **106** of the height sensor assembly(ies) **102** relative to the frame **82** of the header **34**. Moreover, the computing system **210** may be communicatively coupled to the electromagnetic actuator(s) **118** via the communicative link **202**. In this respect, the computing system **210** may be configured to control the operation of the electromagnetic actuator(s) **118**, such as by adjusting magnitude of electromagnetic load applied by the electromagnetic actuator(s) **118**. In addition, the computing system **210** may be communicatively coupled to any other suitable components of the harvester **10** and/or the system **200**.

[0045] In general, the computing system **210** may comprise any suitable processor-based device known in the art, such as a given controller or computing device or any suitable combination of controllers or computing devices. Thus, in several embodiments, the computing system **210** may include one or more processor(s) **212** and associated memory device(s) **214** configured to perform a variety of computer-implemented functions. As used herein, the term “processor” refers not only to integrated circuits referred to in the art as being included in a computer, but also refers to a controller, a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits. Additionally, the memory device(s) **214** of the computing system **210** may generally comprise memory element(s) including, but not limited to, a computer readable medium (e.g., random access memory (RAM)), a computer readable non-volatile medium (e.g., a flash memory), a floppy disc, a compact disc-read only memory (CD-ROM), a magneto-optical disc (MOD), a digital versatile disc (DVD), and/or other suitable memory elements. Such memory device(s) **214** may generally be configured to store suitable computer-readable instructions that, when implemented by the processor(s) **212**, configure the computing system **210** to perform various computer-implemented functions, such as one or more aspects of the methods and algorithms that will be described herein. In addition, the computing system **210** may also include various other suitable components, such as a communications circuit or module, one or more input/output channels, a data/control bus and/or the like.

[0046] It should be appreciated that the computing system **210** may correspond to an existing computing system(s) of the harvester **10**, itself, or the computing system **210** may correspond to a separate processing device. For instance, in one embodiment, the computing system **210** may form all or part of a separate plug-in module that may be installed in association with the harvester **10** to allow for the disclosed systems to be implemented without requiring additional software to be uploaded onto existing control devices of the harvester **10**.

[0047] Furthermore, it should also be appreciated that the functions of the computing system **210** may be performed by a single processor-based device or may be distributed across any number of processor-based devices, in which instance such devices may be considered to form part of the computing system **210**. For instance, the functions of the computing system **210** may be distributed across multiple application-specific controllers or computing devices, such as a navigation controller, an engine computing controller, a transmission controller, an implement controller and/or the like.

[0048] In addition, the system **200** may also include one or more input devices **220**. More specifically, the input device(s) **220** may be configured to receive inputs (e.g., inputs to rotate the sensor arm(s) **106**) from the operator. For example, in several embodiments, the input device(s) **220** may be configured as a user interface(s). The user interface(s) may include a touchscreen(s), keypad(s), touchpad(s), knob(s), button(s), slider(s), switch(s), mice, microphone(s), and/or the like, which are configured to receive inputs from the operator. Such inputs may be used by the computing system **210** for use in controlling the operation of the electromagnetic actuator(s) **118**. Moreover, the user interface(s) may include one or more feedback devices (not shown), such as display screens, speakers, warning lights, and/or the like, which are configured to provide feedback from the computing system **210** to the operator. As such, the input device(s) **220** may, in turn, be communicatively coupled to the computing system **210** via the communicative link **202** to permit the feedback to be transmitted from the computing system **210** to the input device(s) **220**.

[0049] Additionally, the input device(s) **220** may be mounted or otherwise positioned within the operator's cab **30**. However, in alternative embodiments, the input device(s) **220** may mounted at any other suitable location.

[0050] Referring now to FIG. **4**, a flow diagram of one embodiment of control logic **300** that may be executed by the computing system **210** (or any other suitable computing system) for controlling the operation of an agricultural harvester is illustrated in accordance with aspects of the present subject matter. Specifically, the control logic **300** shown in FIG. **4** is representative of steps of one embodiment of an algorithm that can be executed to electronically control the position of the arms of an implement height sensor assembly of a harvesting implement of an agricultural harvester **10**. Thus, in several embodiments, the control logic **300** may be advantageously utilized in association with a system installed on or forming part of an agricultural harvester to allow for real-time control of the harvester without requiring substantial computing resources and/or processing time. However, in other embodiments, the control logic **300** may be used in association with any other suitable system, application, and/or the like for controlling the operation of an agricultural harvester.

[0051] As shown in FIG. **4**, at (**302**), the control logic **300** includes receiving an input to rotate an arm of an implement height sensor assembly to an open position. Specifically, as mentioned above, in several embodiments, the computing system **210** is communicatively coupled to the input device **220** via the communicative link **202**. In this respect, prior to the harvester **10** traveling across the field to perform a harvesting operation thereon, the computing system **210** may receive an operator input from the input device **220** to rotate the sensor arm(s) **106** of the implement height sensor assembly **108** to the open position **88**B relative to the frame **82** of the header **34**.

[0052] Furthermore, at (**304**), the control logic **300** includes controlling an operation of an electromagnetic actuator to rotate the arm to the open position and based on the received input. Specifically, as mentioned above, in several embodiments, the computing system **210** is communicatively coupled to the electromagnetic actuator(s) **118** via the communicative link **202**. In this respect, the computing system **210** is configured to control the operation of the electromagnetic actuator(s) **118**, such as by increasing a magnitude of electromagnetic load applied by the electromagnetic actuator(s) **118**, to rotate the sensor arm(s) **106** to the open position **88**B based on the input received at (**302**).

[0053] Moreover, at (**306**), the control logic **300** includes receiving field condition sensor data indicative of a compaction level of the soil of the field. Specifically, in several embodiments, the computing system **210** is communicatively coupled to the field condition sensor(s) **72** via the communicative link **202**. In this respect, the computing system **210** is configured to receive data from the field condition sensor(s) **72** indicative of the compaction level of the soil of the field.

[0054] Furthermore, at (**308**), the control logic **300** includes determining the compaction level of the soil of the field based on the received field condition sensor data. Specifically, in several embodiments, the computing system **210** is configured to determine the compaction level of the soil of the field based on the data received at (**306**). For example, in one embodiment, the computing system **210** may access a look-up table(s) stored within its memory device(s) **214** that correlates the field condition sensor data received at (**306**) to a value(s) of the compaction level of the soil of the field.

[0055] Additionally, at (**310**), the control logic **300** includes receiving field condition sensor data indicative of a roughness of the surface of the field. Specifically, in several embodiments, the computing system **210** is configured to receive data from the field condition sensor(s) **72** indicative of the roughness of the surface of the field.

[0056] Moreover, at (**312**), the control logic **300** includes determining the roughness of the surface of the field based on the received field condition sensor data. Specifically, in several embodiments, the computing system **210** is configured to determine the roughness of the surface of the field based on the field condition sensor data received at (**310**). For example, in one embodiment, the computing system **210** may access a look-up table(s) stored within its memory device(s) **214** that correlates the field condition sensor data received at (**310**) to a value(s) of the roughness of the surface of the field.

[0057] Furthermore, as shown in FIG. **4**, at (**314**), the control logic **300** includes receiving vehicle speed sensor data indicative of the ground speed of the harvester base vehicle. Specifically, in several embodiments, the computing system **210** is communicatively coupled to the vehicle speed sensor(s) **74** via the communicative link **202**. In this respect, the computing system **210** is configured to receive data from the vehicle speed sensor(s) **74** indicative of the ground speed of the harvester base vehicle **52**.

[0058] Additionally, at (**316**), the control logic **300** includes determining the ground speed of the harvester base vehicle based on the received vehicle speed sensor data. Specifically, in several embodiments, the computing system is configured to determine the ground speed of the harvester base vehicle **52** based on the vehicle speed sensor data received at (**314**).

[0059] Moreover, at (**318**), the control logic **300** includes adjusting a magnitude of electromagnetic load applied by the electromagnetic actuator to prevent the arm from rotating relative to the frame based on at least one of the determined compaction level of the soil, the determined roughness of the surface of the field, and/or the determined ground speed of the harvester base vehicle. Specifically, in several embodiments, the computing system **210** may be configured to adjust the magnitude of the electromagnetic load applied by the electromagnetic actuator(s) **118** to prevent the sensor arm(s) **106** from rotating relative to the frame **82** of the header **34** based on the compaction level of the soil determined at (**308**), the roughness of the surface of the field determined at (**312**), and/or the ground speed of the harvester base vehicle **52** determined at (**316**).

[0060] Referring now to FIG. **5**, a flow diagram of one embodiment of a method **400** for controlling the operation of an agricultural harvester is illustrated in accordance with aspects of the present subject matter. In general, the method **400** will be described herein with reference to the harvester **10** and the system **200** described above with reference to FIGS. **1-4**. However, it should be appreciated by those of ordinary skill in the art that the disclosed method **400** may generally be implemented with any agricultural harvester having any suitable harvester configuration and/or within any system having any suitable system configuration. In addition, although FIG. **5** depicts steps performed in a particular order for purposes of illustration and discussion, the methods discussed herein are not limited to any particular order or arrangement. One skilled in the art, using the disclosures provided herein, will appreciate that various steps of the methods disclosed herein can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure.

[0061] As shown in FIG. **5**, at (**402**), the method **400** includes receiving, with a computing system, an input to rotate an arm of an implement height sensor assembly relative to the harvesting implement between a stored position and an open position in which the arm is configured to contact a surface of the field as the harvesting implement is moved through the field. For instance, as described above, the computing system **210** may be configured to receive an input to rotate the sensor arm(s) **106** of the implement height sensor assembly **102** relative to the header **34** between the stored position **88**A and the open position **88**B.

[0062] Additionally, at (**404**), the method **400** includes controlling, with the computing system, an operation of an electrically activatable actuator coupled to the arm of the implement height sensor assembly to rotate the arm between the stored position and the open position based on the received input and, once in position, prevent the arm from rotating. For instance, as described above, the computing system **210** may be configured to control the operation of the electromagnetic actuator(s) **118** to rotate the sensor arm(s) **106** between the stored position **88**A and the open position **88**B based on the received input.

[0063] It is to be understood that the steps of the control logic **300** and the method **400** are performed by the computing system **210** upon loading and executing software code or instructions which are tangibly stored on a tangible computer readable medium, such as on a magnetic medium, e.g., a computer hard drive, an optical medium, e.g., an optical disc, solid-state memory, e.g., flash memory, or other storage media known in the art. Thus, any of the functionality performed by the computing system **210** described herein, such as the control logic **300** and the method **400**, is implemented in software code or instructions which are tangibly stored on a tangible computer readable medium. The computing system **210** loads the software code or instructions via a direct interface with the computer readable medium or via a wired and/or wireless network. Upon loading and executing such software code or instructions by the computing system **210**, the computing system **210** may perform any of the functionality of the computing system **210** described herein, including any steps of the control logic **300** and the method **400** described herein.

[0064] The term “software code” or “code” used herein refers to any instructions or set of

instructions that influence the operation of a computer or controller. They may exist in a computer-executable form, such as machine code, which is the set of instructions and data directly executed by a computer's central processing unit or by a controller, a human-understandable form, such as source code, which may be compiled in order to be executed by a computer's central processing unit or by a controller, or an intermediate form, such as object code, which is produced by a compiler. As used herein, the term “software code” or “code” also includes any human-understandable computer instructions or set of instructions, e.g., a script, that may be executed on the fly with the aid of an interpreter executed by a computer's central processing unit or by a controller.
[0065] This written description uses examples to disclose the technology, including the best mode, and also to enable any person skilled in the art to practice the technology, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the technology is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A harvesting implement, comprising: a frame configured to be coupled to a harvester base vehicle; a cutter bar configured to sever crop material from a field; a reel rotatably coupled to the frame and configured to direct the crop material toward the cutter bar; an implement height sensor assembly including an arm coupled to the frame, the arm configured to rotate relative to the frame between a stored position and an open position in which the arm is configured to contact a surface of the field as the harvesting implement is moved through the field; a biasing element configured to bias the arm of the implement height sensor toward the surface of the field when the arm is in the open position; and an electrically activatable actuator coupled to the arm of the implement height sensor assembly and configured to rotate the arm between the stored position and the open position and apply a load to prevent the arm from rotating from the stored position or the open position.
2. The harvesting implement of claim 1, wherein the implement height sensor assembly is configured to generate data indicative of a position of the arm of the implement height sensor assembly relative to the frame.
3. The harvesting implement of claim 1, further comprising: an input device configured to receive an operator input to rotate the arm between the stored position and the open position; and a computing system communicatively coupled to the input device, the computing system configured to control the operation of the electrically activatable actuator to rotate the arm based on the received operator input and, once the arm is in position, prevent the arm from rotating relative to the frame.
4. The harvesting implement of claim 1, wherein: the electrically activatable actuator is configured as an electromagnetic actuator configured to apply an electromagnetic load to rotate the arm between the stored position and the open position; and when the arm is in the open position, the electromagnetic load applied by the electromagnetic actuator counteracts a load applied to the arm by the surface of the field to prevent the arm from rotating relative to the frame.
5. The harvesting implement of claim 4, wherein: when the arm is in the stored position, the electromagnetic load applied by the electromagnetic actuator holds the arm in the stored position.
6. The harvesting implement of claim 4, further comprising: a field condition sensor configured to generate data indicative of a compaction level of soil of the field; and a computing system communicatively coupled to the field condition sensor, the computing system configured to: determine the compaction level of the soil of the field based on the data generated by the field condition sensor; and adjust a magnitude of the electromagnetic load applied by the

electromagnetic actuator to prevent the arm from rotating relative to the frame based on the determined compaction level of the soil of the field.

7. The harvesting implement of claim 4, further comprising: a field condition sensor configured to generate data indicative of a roughness of the surface of the field; and a computing system communicatively coupled to the field condition sensor, the computing system configured to: determine the roughness of the surface of the field based on the data generated by the field condition sensor; and adjust a magnitude of the electromagnetic load applied by the electromagnetic actuator to prevent the arm from rotating relative to the frame based on the determined roughness of the surface of the field.

8. The harvesting implement of claim 4, further comprising: a vehicle speed sensor configured to generate data indicative of a ground speed of the harvester base vehicle; and a computing system communicatively coupled to the vehicle speed sensor, the computing system configured to: determine the ground speed of the harvester base vehicle based on the data generated by the field condition sensor; and adjust a magnitude of the electromagnetic load applied by the electromagnetic actuator to prevent the arm from rotating relative to the frame based on the determined ground speed of the harvester base vehicle.

9. The harvesting implement of claim 1, the implement height sensor assembly further comprising: a shaft extending between a first end and a second end, each end coupled to the frame of the harvesting implement, the shaft rotatable about an axis defined between the first end and the second end, and wherein: the arm is fixed to the shaft such that rotation of the shaft results in rotation of the arm between the stored position and the open position, and the electrically activatable actuator is coupled to the shaft and configured to rotate the shaft.

10. A system for controlling the operation of an agricultural harvester, the system comprising: a harvesting implement pivotably coupled to a harvester base vehicle, the harvesting implement configured to be pivoted such that a height of the harvesting implement above a surface of a field is adjusted; an implement height sensor assembly including an arm coupled to the harvesting implement, the arm configured to rotate relative to the harvesting implement between a stored position and an open position in which the arm is configured to contact a surface of the field as the harvesting implement is moved through the field; and an electrically activatable actuator coupled to the arm of the implement height sensor assembly and configured to rotate the arm between the stored position and the open position and apply a load to prevent the arm from rotating from the stored position or the open position; a computing system configured to: receive an input to rotate the arm between the stored position and the open position; and control the operation of the electrically activatable actuator to rotate the arm between the stored position and the open position based on the received input and, once the arm is in position, prevent the arm from rotating.

11. The system of claim 10, wherein the implement height sensor assembly is configured to generate data indicative of a position of the arm of the implement height sensor assembly relative to the frame.

12. The system of claim 10, wherein: the electrically activatable actuator is configured as an electromagnetic actuator configured to apply an electromagnetic load to rotate the arm between the stored position and the open position; and when the arm is in the open position, the electromagnetic load applied by the electromagnetic actuator counteracts a load applied to the arm by the surface of the field to prevent the arm from rotating relative to the harvesting implement.

13. The system of claim 12, wherein: when the arm is in the stored position, the electromagnetic load applied by the electromagnetic actuator holds the arm in the stored position.

14. The system of claim 12, further comprising: a field condition sensor configured to generate data indicative of a compaction level of soil of the field, wherein: the computing system is communicatively coupled to the field condition sensor, the computing system further configured to: determine the compaction level of the soil of the field based on the data generated by the field condition sensor; and adjust a magnitude of the electromagnetic load applied by the electromagnetic actuator to prevent the arm from rotating relative to the harvesting implement based on the determined compaction level of the soil of the field.

15. The system of claim 12, further comprising: a field condition sensor configured to generate data indicative of a roughness of the surface of the field, wherein: the computing system is communicatively coupled to the field condition sensor, the computing system further configured to: determine the roughness of the surface of the field based on the data generated by the field condition sensor; and adjust a magnitude of the electromagnetic load applied by the electromagnetic actuator to prevent the arm from rotating relative to the harvesting implement based on the determined roughness of the surface of the field.

16. The system of claim 12, further comprising: a vehicle speed sensor configured to generate data indicative of a ground speed of the harvester base vehicle, wherein: the computing system is communicatively coupled to the vehicle speed sensor, the computing system further configured to: determine the ground speed of the harvester base vehicle based on the data generated by the field condition sensor; and adjust a magnitude of the electromagnetic load applied by the electromagnetic actuator to prevent the arm from rotating relative to the harvesting implement based on the determined ground speed of the harvester base vehicle.

17. The system of claim 10, the implement height sensor assembly further comprising: a shaft extending between a first end and a second end, each end coupled to the frame of the harvesting implement, the shaft rotatable about an axis defined between the first end and the second end, and wherein: the arm is fixed to the shaft such that rotation of the shaft results in rotation of the arm between the stored position and the open position, and the electrically activatable actuator is coupled to the shaft and configured to rotate the shaft.

18. A method for controlling the operation of an agricultural harvester, the agricultural harvester including a harvesting implement coupled to a harvester base vehicle, the method comprising: receiving, with a computing system, an input to rotate an arm of an implement height sensor assembly relative to the harvesting implement between a stored position and an open position in which the arm is configured to contact a surface of the field as the harvesting implement is moved through the field; and controlling, with the computing system, an operation of an electrically activatable actuator coupled to the arm of the implement height sensor assembly to rotate the arm between the stored position and the open position based on the received input and, once the arm is in position, prevent the arm from rotating.